Tutorial 2

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Similar to HW 2 Problem 3: find the limits

a) $a_n = \ln(3n-6) - \ln(9n)$

$= \ln\left(\frac{3n-6}{9n}\right) = \ln\left(\frac{n-2}{3n}\right) = \ln\left(\frac{1}{3} - \frac{2}{3n}\right)$

$\lim_{n\to\infty} \ln\left(\frac{1}{3} - \frac{2}{3n}\right) = \ln\left(\frac{1}{3}\right) = \ln(1) - \ln(3) = \boxed{-\ln(3)}$

b) $a_n = n\sin(3/n)$

$\lim_{x\to\infty} x\cdot\sin(3/x) = \lim_{n\to\infty} \frac{\sin(3/x)}{1/x} = \lim_{n\to\infty} \frac{\cos(3/x)\cdot -3/x^2}{-1/x^2}$

$= \lim_{x\to\infty} \cos(3/x)(3) = 3$

Recall thm: if $\lim_{x\to\infty} f(x) = L$ and $f(n) = a_n$

then $\lim_{n\to\infty} a_n = L$. Thus $\lim_{n\to\infty} n\sin\left(\frac{3}{n}\right) = 3$

Similar to HW2 Problem 4: converge or not?

(i) $a_n = \frac{(-1)^{n+1} n^2}{n^2-7}$

$\lim_{n\to\infty} \left(\frac{n^2}{n^2-7}\right) = \lim_{n\to\infty} \frac{1}{1-7/n^2} = 1$

So a_n will alternate b/w -1 and 1; diverges.

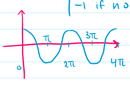
(ii) $a_n = \frac{(-1)^{n+1} n}{n^2-9}$

$\lim_{n\to\infty} |a_n| = \lim_{n\to\infty} \frac{n}{n^2-9} = 0$ converges

Recall: if $\lim_{n\to\infty} |a_n| = 0$ then $\lim_{n\to\infty} a_n = 0$

(iii) $a_n = \cos(n\pi) = \begin{cases} 1 \text{ if } n \text{ even} \\ -1 \text{ if } n \text{ odd} \end{cases}$

note: cos(x)



So $\lim_{n\to\infty} \cos(n\pi)$ diverges.

(iv) $a_n = \sin(n\pi) = 0$ $\lim_{n\to\infty} \sin(n\pi) = \lim_{n\to\infty} 0 = 0$

converges.

Similar to HW2 Problem 5: Evaluate

$\sum_{i=2}^{84} \left(\frac{1}{(i+1)^5} - \frac{1}{(i-1)^5}\right)$ use telescoping

by some terms: $\left(\frac{1}{(2+1)^5} - \frac{1}{(2-1)^5}\right) + \left(\frac{1}{(3+1)^5} - \frac{1}{(3-1)^5}\right) + \left(\frac{1}{(4+1)^5} - \frac{1}{(4-1)^5}\right) + \left(\frac{1}{(5+1)^5} - \frac{1}{(5-1)^5}\right)$

$+\ldots+ \left(\frac{1}{(81+1)^5} - \frac{1}{(81-1)^5}\right) + \left(\frac{1}{(82+1)^5} - \frac{1}{(82-1)^5}\right) + \left(\frac{1}{(83+1)^5} - \frac{1}{(83-1)^5}\right) + \left(\frac{1}{(84+1)^5} - \frac{1}{(84-1)^5}\right)$

$\left(\frac{1}{3^5} - \frac{1}{1^5}\right) + \left(\frac{1}{4^5} - \frac{1}{2^5}\right) + \left(\frac{1}{5^5} - \frac{1}{3^5}\right) + \left(\frac{1}{6^5} - \frac{1}{4^5}\right)$

$+\ldots+ \left(\frac{1}{82^5} - \frac{1}{80^5}\right) + \left(\frac{1}{83^5} - \frac{1}{81^5}\right) + \left(\frac{1}{84^5} - \frac{1}{82^5}\right) + \left(\frac{1}{85^5} - \frac{1}{83^5}\right)$

$= \boxed{-1 - \frac{1}{2^5} + \frac{1}{84^5} + \frac{1}{85^5}}$ ← can write this as final answer.

Similar to HW2 Problem 6: Find the sum

a) $10 + \underbrace{\frac{5}{2}}_{\times\frac{1}{4}} + \underbrace{\frac{5}{8}}_{\times\frac{1}{4}} + \underbrace{\frac{5}{32}}_{\times\frac{1}{4}} + \underbrace{\frac{5}{128}}_{\times\frac{1}{4}} + \underbrace{\frac{5}{512}}_{\times\frac{1}{4}} + \underbrace{\frac{5}{2048}}_{\times\frac{1}{4}} + \ldots$

$= 5\left(2 + \frac{1}{2} + \frac{1}{8} + \frac{1}{32} + \ldots\right)$

$= 5\sum_{k=0}^{\infty} \frac{2}{4^k}$ turn to geometric:

$= 5\sum_{k=1}^{\infty} 2\cdot 4^{k-1} = \sum_{k=1}^{\infty} 10\cdot\left(\frac{1}{4}\right)^{k-1}$ $a = 10$, $r = 1/4$

sum $= \frac{10}{1-1/4} = \frac{40}{3}$

b) $\sum_{n=2}^{\infty} \frac{3}{n(n+3)}$ note that $\frac{3}{n(n+3)} = \frac{A}{n} + \frac{B}{n+3}$

$\Rightarrow 3 = A(n+3) + Bn \Rightarrow 3 = An + 3A + Bn$

$\Rightarrow 3 = 3A$ and $0 = A + B$

$A = 1$, $B = -1 \Rightarrow \sum_{n=2}^{\infty} \left(\frac{1}{n} - \frac{1}{(n+3)}\right)$

telescoping: $\left(\frac{1}{2} - \frac{1}{5}\right) + \left(\frac{1}{3} - \frac{1}{6}\right) + \left(\frac{1}{4} - \frac{1}{7}\right) + \left(\frac{1}{5} - \frac{1}{8}\right)$

$= \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12}$

Similar to HW3 Problem 2: use integral test to see if it converges or diverges.

(i) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^4}$ $\int_2^{\infty} \frac{1}{x(\ln x)^4}\,dx = \int_{\ln(2)}^{\infty} u^{-4}\,du = \frac{u^{-3}}{-3} + C \Big|_{\ln(2)}^{\infty}$

let $u = \ln(x)$ $du = \frac{1}{x}dx$ $= \lim_{u\to\infty} \frac{1}{3u^3} + C + \frac{1}{3\ln^3(2)} - C$

as $x\to\infty$ $u\to\infty$, $x\to 2$, $u\to\ln(2)$ $= \frac{1}{3\ln^3(2)}$ which converges.

$\therefore$ this series converges.

(ii) $\sum_{n=1}^{\infty} n^5 e^{n^6}$ $u = x^6$, $du = 6x^5dx$ $x\to\infty, u\to\infty$; $x\to 1, u\to 1$

$\int_1^{\infty} x^5 e^{x^6}\,dx = \int_1^{\infty} \frac{1}{6} e^u\,du = \frac{1}{6} e^u + C \Big|_1^{\infty}$

$= \underbrace{\lim_{u\to\infty} \frac{1}{6} e^u + C}_{\infty} - \frac{1}{6} e^{(1)} - C$ diverges

$\therefore$ This series diverges.

(iii) $\sum_{n=2}^{\infty} \frac{\ln(n)}{n^{1/2}}$ $\int_2^{\infty} \frac{\ln(x)}{x^{1/2}}\,dx$

use: integration by parts: $u = \ln(x)$ $dv = x^{-1/2}dx$

$uv - \int v\,du$ $du = \frac{1}{x}dx$ $v = \frac{x^{1/2}}{1/2}$

$2\ln(x)x^{1/2} - \int 2x^{1/2}x^{-1}\,dx$

$= 2\ln(x)x^{1/2} - 2\int x^{-1/2}\,dx$

$= 2\ln(x)x^{1/2} - 2\left(\frac{x^{1/2}}{1/2}\right)$

$= 2\ln(x)x^{1/2} - 4x^{1/2} = 2x^{1/2}(\ln(x) - 2) + C \Big|_2^{\infty}$

$= \underbrace{\lim_{x\to\infty} 2x^{1/2}(\ln(x) - 2)}_{\infty} - 2(2)^{1/2}(\ln(2) - 2))$

divergent $\Rightarrow$ this series diverges.